MultiCore TECHNOLOGIES

LEARN TODAY BUILD TOMORROW

Your Future in Tech

DATA SCIENCE WITH GEN AI

From Data to Intelligent Solutions



COURSE HIGHLIGHTS

- Python & Data Science
- Machine Learning & Artificial Intelligence
- Deep Learning & Neural Networks
- SQL & Power BI
- Generative AI, LLMs, Prompt Engineering & RAG

DATA SCIENCE WITH AI & GENERATIVE AI

From Data to Intelligent Solutions

COURSE OVERVIEW

Build a strong foundation in Data Science, Artificial Intelligence and Generative AI through a structured learning journey from Python programming to advanced AI applications.

This program covers data analysis, machine learning, deep learning, SQL, Power BI, NLP, computer vision, Generative AI, LLMs, Prompt Engineering and RAG.

WHAT YOU WILL LEARN

1. **Python & Advanced Python**
   - Programming fundamentals
   - Python libraries
2. **Data Science with Python**
   - Data cleaning & transformation
   - Analysis & visualization
3. **Machine Learning**
   - Supervised & unsupervised learning
   - Predictive models
4. **Artificial Intelligence**
   - Intelligent systems
   - NLP & computer vision
5. **Deep Learning**
   - Neural networks
   - Advanced architectures
6. **SQL & Database Concepts**
   - Querying data
   - Data management
7. **Power BI**
   - Dashboards
   - Business intelligence reports
8. **Generative AI**
   - LLMs & Transformers
   - Prompt Engineering & RAG

WHO CAN JOIN?

- Students
- Graduates
- Beginners
- Working Professionals
- Developers
- Aspiring Data & AI Professionals

LEARNING OUTCOME

By the end of the program, learners will be able to analyze data, build ML models, develop AI solutions, create dashboards and work with modern Generative AI technologies.

SKILLS TODAY
FOR A SMARTER TOMORROW

DATA SCIENCE & MACHINE LEARNING

PYTHON | DATA SCIENCE | ML

1 PYTHON & ADVANCED PYTHON

PYTHON FUNDAMENTALS

- Variables, Data Types & Operators
- Conditional Statements
- For & While Loops
- Functions & Modules
- Exception Handling
- File Handling
- Object-Oriented Programming

PYTHON LIBRARIES

NumPy
Arrays, Matrix Operations & Vectorized Computations

Pandas
Series, DataFrames & Data Manipulation

Matplotlib & Seaborn
Data Visualization & Statistical Plots

SciPy
Scientific Computing, Optimization & Linear Algebra

2 DATA SCIENCE WITH PYTHON

DATA CLEANING & PREPROCESSING

- Missing Data Handling
- Data Imputation
- Data Transformation
- Normalization & Scaling
- Categorical Data Encoding
- Outlier Detection & Handling

EXPLORATORY & ANALYSIS

- Descriptive Statistics
- Correlation Analysis
- Distribution Analysis
- Histograms & Box Plots
- Data Aggregation & Grouping

DATA MANIPULATION

- DataFrame Operations
- Data Merging & Joining
- Concatenation
- One-to-One Relationships
- One-to-Many Relationships
- Many-to-Many Relationships

3 MACHINE LEARNING

SUPERVISED LEARNING

- Linear Regression
- Polynomial Regression
- Logistic Regression
- K-Nearest Neighbors
- Decision Trees
- Support Vector Machines
- Random Forests

UNSUPERVISED LEARNING

- K-Means Clustering
- Hierarchical Clustering
- PCA
- t-SNE
- Apriori
- Eclat

MODEL EVALUATION

- Accuracy
- Precision & Recall
- F1-Score
- ROC & AUC
- Cross-Validation
- Grid Search
- Hyperparameter Tuning
- Overfitting & Underfitting

AI

BUILD INTELLIGENT SOLUTIONS

ANALYZE REAL-WORLD DATA

WORK WITH MODERN TOOLS

CREATE DATA-DRIVEN IMPACT

SKILLS TODAY
FOR A SMARTER TOMORROW

DATA SCIENCE & AI, DEEP LEARNING, SQL & POWER BI

Build Smarter Solutions with Data & Gen AI

ARTIFICIAL INTELLIGENCE & DEEP LEARNING

4 ARTIFICIAL INTELLIGENCE

AI Fundamentals

- Introduction to AI
- Problem Solving
- Search Algorithms
- A* Algorithm
- Minimax
- Alpha-Beta Pruning

NATURAL LANGUAGE PROCESSING

- Text Preprocessing
- Tokenization
- Lemmatization
- Stopword Removal
- Text Classification
- Sentiment Analysis
- Named Entity Recognition

COMPUTER VISION

- Image Classification
- Object Detection
- Convolutional Neural Networks

5 DEEP LEARNING

Neural Networks

- Perceptron
- Activation Functions
- Backpropagation
- Gradient Descent

Deep Learning Models

- Feedforward Neural Networks
- Convolutional Neural Networks
- Recurrent Neural Networks
- Generative Adversarial Networks
- Autoencoders

Frameworks & Advanced Learning

- TensorFlow & Keras
- PyTorch
- Transfer Learning
- Model Fine-Tuning



BUSINESS INTELLIGENCE & DATABASES

6 POWER BI

- Power BI Desktop
- Data Connectivity
- Power Query
- Data Transformation
- Data Modeling
- Reports & Dashboards
- Interactive Visualizations
- Power BI Service
- Publishing & Sharing

7 SQL

SQL Fundamentals

- SELECT & WHERE
- ORDER BY & DISTINCT
- COUNT, SUM & AVG
- MIN & MAX

Advanced SQL

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL JOIN
- GROUP BY & HAVING
- Subqueries & Nested Queries
- Window Functions

Database Concepts

- CASE Statements
- GROUPING SETS
- ROLLUP & CUBE
- Normalization & Denormalization
- Indexing
- Views & Triggers

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SKILLS TODAY
FOR A SMARTER TOMORROW

DATA SCIENCE & GENERATIVE AI

BUILD INTELLIGENT SOLUTIONS FOR A SMARTER TOMORROW

8 FOUNDATIONS OF AI & GENERATIVE AI

8 AI FUNDAMENTALS

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Relationship Between AI, ML & DL

GENERATIVE AI

- Introduction to Generative AI
- Traditional AI vs Generative AI
- How Generative AI Works
- Real-World Applications

GEN AI APPLICATIONS

AI Chatbots
Customer Support, Query Resolution & Conversational AI

Content Generation
Emails, Reports, Articles & Marketing Content

Image Generation
Creating Visual Content from Text Prompts

Code Generation
Code Creation, Debugging & Development Assistance



9 TRANSFORMERS & LARGE LANGUAGE MODELS

9 TRANSFORMER ARCHITECTURE

- Transformer Fundamentals
- Sequential Data Processing
- Encoder & Decoder Concepts
- Context Understanding

ATTENTION MECHANISM

- Self-Attention
- Contextual Understanding
- Token Importance
- Attention-Based Processing

LARGE LANGUAGE MODELS

- LLM Fundamentals
- GPT
- BERT
- Text Generation
- Language Understanding

10 PROMPT ENGINEERING

EFFECTIVE PROMPTING

- Writing Clear & Specific Prompts
- Defining Context & Instructions
- Setting Constraints
- Controlling Output Format

PROMPTING TECHNIQUES

- Zero-Shot Prompting
- Few-Shot Prompting
- Role-Based Prompting
- Structured Prompting
- Step-by-Step Problem Solving

BUILD INTELLIGENT SOLUTIONS

ANALYZE REAL-WORLD DATA

WORK WITH MODERN TOOLS

CREATE DATA-DRIVEN IMPACT

SKILLS TODAY
FOR A SMARTER TOMORROW

DATA SCIENCE & GENERATIVE AI

RAG, PROJECTS & CAREER

11 RETRIEVAL AUGMENTED GENERATION — RAG

RAG FUNDAMENTALS

- Introduction to RAG
- Information Retrieval
- Knowledge-Based AI
- Retrieval & Generation
- Grounding AI Responses

RAG WORKFLOW

USER QUERY

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RETRIEVE RELEVANT INFORMATION

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AUGMENT THE PROMPT

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GENERATE RESPONSE

RAG APPLICATIONS

- Document Question Answering
- Knowledge Assistants
- Enterprise Chatbots
- Customer Support Systems
- Internal Knowledge Search

12 HANDS-ON PROJECTS

BUILD REAL-WORLD AI & DATA PROJECTS

Data Analytics Dashboard

Analyze business data and create interactive Power BI dashboards.

Machine Learning Prediction System

Build and evaluate predictive machine learning models.

AI Chatbot

Develop an intelligent conversational AI application.

RAG-Based Knowledge Assistant

Build an AI application capable of answering questions from documents and knowledge sources.

Data Science Project

Perform data cleaning, analysis, visualization and predictive modeling.



BUILD INTELLIGENT SOLUTIONS

ANALYZE REAL-WORLD DATA

WORK WITH MODERN TOOLS

CREATE DATA-DRIVEN IMPACT

SKILLS TODAY
FOR A SMARTER TOMORROW

13 SKILLS YOU WILL MASTER

- Python
- Data Science
- SQL
- Power BI
- Machine Learning
- Artificial Intelligence
- Deep Learning
- NLP
- Computer Vision
- Generative AI
- LLMs
- Prompt Engineering
- Transformers
- RAG

14 CAREER OPPORTUNITIES

- Data Analyst
- Data Scientist
- Generative AI Developer
- AI/ML Developer
- Machine Learning Engineer
- AI Engineer
- Business Intelligence Analyst
- Data & AI Consultant